

**REAL-TIME OBJECT DETECTION AND COOPERATIVE
NAVIGATION IN SWARM ROBOTICS USING
EMBEDDED EDGE AI**

PROJECT-21ECP302L

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SRM INSTITUTE OF SCIENCE AND TECHNOLOGY



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BONAFIDE CERTIFICATE

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DECLARATION

We hereby declare that the Project entitled “REAL-TIME OBJECT DETECTION AND COOPERATIVE NAVIGATION IN SWARM RPBOTS USING EMBEDDED EDGE AI ” to be submitted for the course “**21ECP302L – Project**”, is our original work as a team.

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ABSTRACT

Swarm robotics is an emerging field inspired by the collective behavior of natural systems such as ants and bees, where multiple simple agents collaborate to accomplish complex tasks. This project presents the design and implementation of a low-cost swarm robotic system using ESP32 module. The system follows a master–slave architecture in which a central controller coordinates multiple robotic units through wireless communication.

Each robot in the swarm is equipped with essential components such as microcontrollers, motor drivers, sensors, and communication modules to perform tasks like navigation, obstacle avoidance, and object detection. The robots cooperate by sharing information, enabling efficient task distribution and improved performance compared to individual robots.

SUSTAINABLE DEVELOPMENT GOALS:

SDG 9 - Industry, Innovation and Infrastructure Supports development of advanced robotic automation and intelligent embedded systems. Swarm robots enhance smart automation and build efficient, intelligent infrastructure systems.



SDG 16 - Peace, Justice and Strong Institutions Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels. Swarm robots can support surveillance, disaster response, and security for safer and more organized societies.



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ABBREVIATIONS

S.NO	ABBREVIATION	FULL FORM
1	IoT	Internet of Things
2	CAM	Camera Module
3	AI	Artificial Intelligence
4	ESP32 AI Thinker Module	ESP32 AI Thinker Module
5	LED	Light Emitting Diode
6	SDG	Sustainable Development Goal
7	Wi Fi	Wireless Fidelity
8	PWM	Pulse Width Modulation
9	RTS	Request To Send
10	CTS	Clear To Send
11	IDE	Integrated Development Environment
12	CPU	Central Processing Unit
13	CSI	Channel State Information
14	GPIO	General Purpose Input Output
15	IDE	Integrated Development Environment
16	RAM	Random Access Memory

17	ACM	Association for Computing Machinery
18	IEEE	Institute of Electrical and Electronics Engineers
19	API	Application Programmable Interface
20	GHz	Gigahertz
21	MHz	Megahertz
22	MCU	Micro Controller Unit
23	Vcc	Supply Voltage

CHAPTER 1

INTRODUCTION

1.1 OBJECTIVE

The primary objective of this project is to design and develop a low-cost swarm robotic system using the ESP32 that enables multiple robots to communicate and coordinate effectively. The system aims to implement a master–slave architecture in which one robot performs decision-making and control, while the remaining robots execute tasks based on received instructions. This approach simplifies coordination while maintaining efficient distributed operation within the swarm.

Another key objective is to enable cooperative task execution such as object detection, navigation, and formation control. The project also focuses on integrating basic vision-based techniques using camera modules to enhance environmental awareness and object recognition capabilities. Additionally, reliable wireless communication is established using built-in Wi-Fi and Bluetooth features of the ESP32, ensuring real-time data exchange and synchronization among robots.

The project further aims to develop a scalable and modular system that allows easy expansion by adding more robots without major design changes. It also emphasizes implementing simple task allocation strategies to distribute work efficiently among robots. Energy efficiency is considered an important objective, with the use of optimized hardware and control techniques to extend operational time. Overall, the project seeks to create a strong foundation for future advancements such as artificial intelligence, IoT integration, and fully autonomous swarm behavior.

In addition, the project aims to improve system robustness by incorporating basic error-handling and fault-tolerance mechanisms during communication and operation. It also seeks to enhance the overall autonomy of the swarm by enabling adaptive behavior based on environmental conditions and sensor inputs.

Overall, the project seeks to create a strong foundation for future advancements such as artificial intelligence, IoT integration, and fully autonomous swarm behavior.

1.2 IMPORTANCE OF PROJECT

This project is important as it demonstrates the practical implementation of swarm robotics, a field that emphasizes decentralized control and collective intelligence. Unlike traditional centralized systems, swarm robotics improves reliability and fault tolerance, as the failure of one robot does not significantly affect the overall system. This makes it highly suitable for real-world applications that require robustness and adaptability.

Another major significance of the project lies in its cost-effectiveness. By utilizing affordable components like the ESP32, the system becomes accessible for educational institutions and research purposes. It provides a valuable platform for students to gain hands-on experience in embedded systems, wireless communication, and multi-robot coordination.

Furthermore, the project has wide-ranging applications in areas such as search and rescue operations, agriculture, environmental monitoring, and industrial automation. Its scalable and flexible design allows it to adapt to different task requirements, making it highly versatile. The project also supports the development of future technologies by serving as a base for integrating advanced concepts like artificial intelligence, Internet of Things (IoT), and smart automation systems.

In addition, the project highlights the importance of energy efficiency and optimized system design in modern robotics. It encourages interdisciplinary learning by combining principles from electronics, computer science, and mechanical engineering. Overall, this project contributes to the advancement of intelligent automation and aligns with current technological trends, making it highly relevant in today's rapidly evolving engineering landscape.

Moreover, the project contributes to improving communication strategies in multi-agent systems, which is essential for achieving coordinated behavior in complex environments. It also helps in understanding practical challenges such as signal interference, synchronization issues, and hardware limitations.

CHAPTER 2

LITERATURE SURVEY

[1]: Real-Time Object Detection on Embedded Edge Devices Using YOLO-Based Models

Authors: J. Redmon et al.

Year: 2024

Redmon et al. (2024) presented a real-time object detection system using lightweight YOLO models optimized for embedded edge devices. The system processes camera input locally, eliminating cloud dependency and enabling low-latency detection of objects such as obstacles and targets. Model optimization techniques like pruning and quantization reduce computational complexity while maintaining acceptable accuracy. The system performs efficiently on resource-constrained hardware, though accuracy may decrease compared to full-scale models and performance can be affected in complex or low-light environments.

Paper 2: Cooperative Navigation in Swarm Robotics Using Distributed Algorithms

Authors: M. Dorigo et al.

Year: 2024

Dorigo et al. (2024) proposed a distributed control algorithm for cooperative navigation in swarm robotics, where multiple robots operate without centralized control by sharing local information with neighboring units. The system is inspired by swarm intelligence principles such as flocking and collective decision-making, enabling scalability and robustness. However, communication overhead between robots can increase, and convergence speed may be slower in complex or dynamic environments.

Paper 3: Edge AI-Based Vision System for Autonomous Mobile Robots

Authors: S. Liu et al.

Year: 2024

Liu et al. (2024) developed an edge AI-based vision system that integrates deep learning models for real-time object detection and classification on mobile robots. The system utilizes onboard processing platforms such as embedded GPUs to perform inference locally, allowing dynamic path adjustment based on detected obstacles. While the approach enables fast and autonomous decision-making, it involves higher power consumption and increased hardware cost.

Paper 4: Multi-Robot Cooperative Exploration Using SLAM Techniques

Authors: H. Zhang et al.

Year: 2023

Zhang et al. (2023) proposed a cooperative exploration framework using SLAM, where multiple robots collaboratively build and share environmental maps to improve navigation efficiency. The system reduces redundancy and enhances coverage by exchanging positional and mapping data among robots. However, synchronization and data consistency between robots increase computational complexity and system overhead.

Paper 5: Lightweight Deep Learning Models for Embedded Vision Systems

Authors: A. Howard et al.

Year: 2023

Howard et al. (2023) introduced lightweight convolutional neural network architectures such as MobileNet designed for embedded vision applications. These models are optimized for low-power devices and provide efficient object detection with reduced computational requirements. Although suitable for real-time applications, they may exhibit lower accuracy and limited feature extraction capability compared to larger models.

Paper 6: Swarm Robotics Communication Using IoT and Wireless Networks

Authors: K. Kumar et al.

Year: 2023

Kumar et al. (2023) proposed a communication framework for swarm robotics based on IoT and wireless networking, enabling robots to exchange data such as position, detected objects, and navigation paths in real time. This enhances coordination and scalability of the swarm, but introduces challenges such as network latency, communication failures, and reliability issues in dynamic environments. The system utilizes common wireless protocols to establish a decentralized communication network, allowing each robot to act as both a sender and receiver of information. This improves coordination, collective decision-making, and adaptability of the swarm in dynamic environments. The framework also supports scalability by allowing additional robots to join the network without significant changes to the architecture. However, challenges such as network latency, packet loss, bandwidth limitations, and communication reliability can affect overall system performance, especially in large-scale deployments or environments with interference.

Paper 7: Autonomous Obstacle Avoidance Using Computer Vision and Sensors

Authors: R. Patel et al.

Year: 2023

Patel et al. (2023) developed an obstacle avoidance system that combines computer vision with sensor inputs such as ultrasonic and LiDAR for improved detection accuracy. The system enables robots to navigate safely in dynamic environments by integrating multiple sensing modalities, although it increases hardware complexity and overall system cost.

Paper 8: Distributed Edge Computing for Multi-Robot Systems

Authors: Y. Chen et al.

Year: 2022

Chen et al. (2022) introduced a distributed edge computing framework in which each robot processes data locally and shares only essential information with others. This reduces network dependency and improves system efficiency while maintaining real-time performance. However, individual robots have limited processing capability, and efficient task allocation becomes necessary for optimal performance.

Paper 9: Reinforcement Learning for Cooperative Robot Navigation

Authors: P. Singh et al.

Year: 2022

Singh et al. (2022) applied reinforcement learning techniques to enable cooperative navigation in swarm robotics, where robots learn optimal behaviors through interaction with the environment and other agents. This approach improves adaptability and decision-making over time but requires large datasets and significant training time, making real-time implementation challenging. Basic swarm behaviors like aggregation, dispersion, and path following are implemented using rule-based algorithms. While the system demonstrates effective coordination at low cost, it is limited by reduced processing power, lack of advanced sensing capabilities, and inability to support complex real-time tasks such as high-accuracy object detection or dynamic decision-making in complex environments. Additionally, scalability to larger swarms may introduce communication and synchronization challenges, requiring more optimized protocols and processing capabilities.

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CHAPTER 3

SYSTEM DESCRIPTION AND METHODOLOGY

3.1 SYSTEM OVERVIEW

The proposed system is a swarm robotic platform designed to achieve coordinated operation among multiple robots using wireless communication and simple control strategies. The system is built using the ESP32, which serves as the main controller for each robotic unit due to its processing capability and built-in communication features.

The architecture of the system follows a master–slave model, where one robot acts as the master unit and the remaining robots function as slave units. The master robot is responsible for controlling the overall operation of the swarm by sending commands to the slave robots. Based on these commands, the slave robots perform specific tasks such as movement, object tracking, or maintaining a formation. This structure simplifies coordination and ensures efficient execution of tasks.

Each robot in the system consists of essential hardware components including a microcontroller, motor driver, DC motors, power supply, and basic sensors. The motor driver controls the movement of the robot by regulating motor speed and direction, while the sensors help in detecting obstacles and assisting navigation. The compact and modular design of each robot allows easy integration and scalability of the system.

Communication between the robots is established using the wireless capabilities of the ESP32. The master robot transmits instructions to the slave robots, enabling synchronized operation. The communication system is designed to be simple and reliable, ensuring that commands are received and executed without significant delay.

The system is capable of performing coordinated tasks through predefined control logic. The robots can move together, follow a leader, or perform basic cooperative actions based on the instructions received from the master unit. This demonstrates the fundamental concept of swarm robotics, where multiple robots work together to achieve a common goal.

Overall, the system provides a simple, cost-effective, and scalable solution for implementing swarm robotic behavior. It demonstrates how multiple robots can communicate, coordinate, and perform tasks collectively using basic hardware and efficient control mechanisms.

In addition to basic coordination, the system is designed with flexibility to support future enhancements and advanced functionalities. The use of the ESP32 allows integration with additional modules such as cameras, advanced sensors, or Internet-based control systems. This makes the platform suitable not only for demonstrating fundamental swarm behavior but also for extending into more complex applications such as autonomous navigation and intelligent decision-making. The overall design ensures that the system remains adaptable, making it a strong foundation for further research and development in swarm robotics.

3.2 BLOCK DIAGRAM

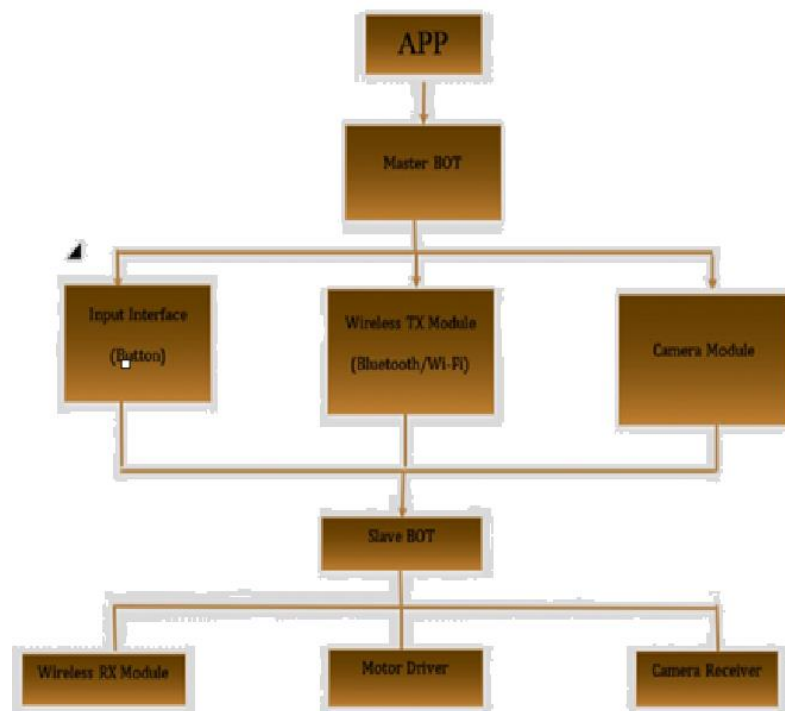


Figure 3.2 (1)Block Diagram

The proposed system is a wireless-controlled robotic architecture designed to enable efficient communication and coordination between robotic units. It is based on a master–slave configuration, where one robot controls and monitors the operation of another robot remotely. This approach improves flexibility and reduces human intervention in critical tasks.

The system begins with a mobile application that acts as the primary control interface. Through this application, the user can send commands to the master bot, allowing easy and intuitive operation. This eliminates the need for complex manual control mechanisms.

The master bot serves as the central processing unit of the system. It receives input from the mobile application and processes the commands accordingly. This bot plays a crucial role in decision-making and signal transmission.

In addition to app-based control, the system also includes an input interface in the form of push buttons. These buttons allow manual operation of the robot when required, ensuring reliability even in the absence of a mobile connection.

A wireless transmission module is integrated into the master bot for communication purposes. Technologies such as Bluetooth or Wi-Fi are used to establish a stable connection between the master and slave bots. This ensures fast and accurate data transfer.

The master bot is also equipped with a camera module to capture real-time visual data. This feature enables the user to monitor the surrounding environment remotely, enhancing the system's functionality.

The captured video data is transmitted along with control signals to the slave bot. This allows the system to provide both control and feedback simultaneously, improving operational efficiency.

The slave bot acts as the execution unit of the system. It receives commands from the master bot and performs actions accordingly. This separation of control and execution simplifies the overall design.

A wireless receiver module is used in the slave bot to receive signals from the master bot. This module ensures that communication is maintained without significant delay or data loss.

3.3 HARDWARE

3.3.1. Microcontroller & Processing Unit

- ESP32-CAM Module (AI Thinker) ×2
- Used in both master and follower robots
- Provides WiFi connectivity, onboard processing, and camera interface
- Handles streaming, communication, and control logic

3.3.2. Motor Driver Module

- L298N Dual H-Bridge Motor Driver ×2
- Controls direction and speed of DC motors
- Interfaces between ESP32 and motors

3.3.3. Locomotion System (Motors & Wheels)

- DC Geared Motors (BO Motors)
- 4 motors (Master robot – 4WD)
- 2 motors (Follower robot – 2WD)
- Robot Wheels
- Robot Chassis (for both bots)

3.3.4. Sensors

- HC-SR04 Ultrasonic Sensor ×2
- Used for obstacle detection
- Measures distance using echo timing

3.3.5. Power Supply System

- 18650 Li-ion Batteries
- Battery Holder
- XL6009E1 Boost Converter Module

3.3.6. Camera & Vision

- OV2640 Camera (integrated with ESP32-CAM)
- Used for live streaming
- Used for object detection (colour-based detection)

3.3.7. Communication & Networking

- Wi-Fi (ESP32 Built-in)
- Enables:
 - Live video streaming
 - HTTP communication between robots
 - Web interface hosting

3.3.8. Software & Development Tools

- Arduino IDE / Embedded C (for ESP32 programming)
- HTML, CSS, JavaScript (for web interface)
- HTTP Protocol (REST-like communication)
- Image Processing (Colour Segmentation Logic)

3.4 STEP-BY-STEP APPROACH FOR DESIGN AND DEVELOPMENT

3.4.1.RequirementAnalysis

The first step involves identifying the system requirements such as remote control capability, real-time communication, video transmission, and reliable robot movement. The objectives and constraints of the system are clearly defined at this stage.

3.4.2. System Design Planning

A complete architecture is designed using a master–slave model. The roles of the master bot, slave bot, mobile application, and communication modules are planned to ensure proper system integration.

3.4.3. Hardware Selection

Suitable hardware components such as microcontrollers, motor drivers, wireless modules (Bluetooth/Wi-Fi), cameras, and power supply units are selected based on performance, cost, and compatibility.

3.4.4. Mobile Application Development

A mobile application is developed to send control commands to the master bot. The interface is designed to be simple and user-friendly for real-time robot control.

3.4.5. Master Bot Development

The master bot is programmed to receive input from the mobile app and process it. It acts as the central control unit responsible for managing communication and command distribution.

3.4.6. Wireless Communication Setup

A wireless communication system using Bluetooth or Wi-Fi is established between the master bot and slave bot. This ensures smooth and continuous data transmission.

3.4.7. Camera Integration

A camera module is integrated into the master bot to capture real-time video. The video stream is configured to be transmitted for monitoring and analysis purposes.

3.4.8. Slave Bot Development

The slave bot is built to execute commands received from the master bot. It is responsible for physical movement and task execution in the environment.

3.4.9. Motor Control Implementation

A motor driver circuit is implemented to control the direction and speed of the slave bot. This ensures accurate movement based on received instructions.

3.4.10. Testing and Integration

Finally, all components are integrated and tested as a complete system. Functional testing is conducted to ensure proper communication, movement accuracy, and real-time response.

3.4.11. System Calibration and Optimization

After initial integration, the system is calibrated to improve accuracy and efficiency. Parameters such as motor speed, signal delay, and wireless transmission range are fine-tuned for smooth performance.

3.4.12. Error Detection and Debugging

The system is tested for possible faults such as communication failure, incorrect command execution, or hardware mismatch. Debugging is carried out in both software and hardware modules to ensure stability.

3.4.13. Power Supply Optimization

A stable and efficient power distribution system is implemented for both master and slave bots.

3.5 TOOLS AND SOFTWARE USED

Arduino IDE

The Arduino Integrated Development Environment (IDE) is used to write, compile, and upload code to microcontrollers such as ESP32. It provides a simple interface for developing embedded applications and supports various libraries required for sensor interfacing and communication.

Proteus Design Suite

Proteus Design Suite is used for circuit design and simulation. It allows virtual testing of electronic components and embedded systems before hardware implementation, helping to verify the functionality and reduce design errors.

Program:**Master:**

```
#include "soc/soc.h"
```

```
#include "soc/rtc_cntl_reg.h"
```

```
#include "esp_camera.h"
```

```
#include <WiFi.h>
```

```
#include "esp_http_server.h"
```

```
#include "img_converters.h"
```

```
#define PWDN_GPIO_NUM 32
```

```
#define RESET_GPIO_NUM -1
```

```
#define XCLK_GPIO_NUM 0
```

```
#define SIOD_GPIO_NUM 26
```

```
#define SIOC_GPIO_NUM 27
```

```
#define Y9_GPIO_NUM 35
```

```
#define Y8_GPIO_NUM 34
```

```
#define Y7_GPIO_NUM 39
```

```
#define Y6_GPIO_NUM 36
```

```
#define Y5_GPIO_NUM 21
```

```
#define Y4_GPIO_NUM 19
```

```
#define Y3_GPIO_NUM 18
```

```
#define Y2_GPIO_NUM 5
```

```
#define VSYNC_GPIO_NUM 25

#define HREF_GPIO_NUM 23

#define PCLK_GPIO_NUM 22


#define IN1 12

#define IN2 13

#define IN3 15

#define IN4 14

#define TRIG_PIN 2

#define ECHO_PIN 4


const char* ssid = "abhi";

const char* password = "abhiram123";


httpd_handle_t server = NULL;

float distance = 999;

unsigned long lastDistanceCheck = 0;

unsigned long stateStartTime = 0;

int robotState = 0;

String slaveIP = "";

String slaveStatus = "Waiting";

String masterStatus = "Running";
```

SLAVE:

```
#include "soc/soc.h"
```

```
#include "soc/rtc_cntl_reg.h"
```

```
#include "esp_camera.h"
```

```
#include <WiFi.h>
```

```
#include "esp_http_server.h"
```

```
#include "img_converters.h"
```

```
#define PWDN_GPIO_NUM 32
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```
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```

```
#define XCLK_GPIO_NUM 0
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#define Y9_GPIO_NUM 35
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```

```
#define Y7_GPIO_NUM 39
```

```
#define Y6_GPIO_NUM 36
```

```
#define Y5_GPIO_NUM 21
```

```
#define Y4_GPIO_NUM 19
```

```
#define Y3_GPIO_NUM 18
```

```
#define Y2_GPIO_NUM 5
```

```
#define VSYNC_GPIO_NUM 25
```

```

#define HREF_GPIO_NUM 23

html += "setInterval(updateStatus,1000);";

html += "</script></body></html>";

httpd_resp_set_type(req, "text/html");

httpd_resp_send(req, html.c_str(), html.length());

return ESP_OK;

}

static esp_err_t status_handler(httpd_req_t *req) {

char json[150];

snprintf(json, sizeof(json),

"{'status': '%s', 'distance': '%.1f', 'wifi': '%s'}",

slaveStatus.c_str(), distance,

WiFi.status() == WL_CONNECTED ? "Connected" : "Reconnecting...");

httpd_resp_set_type(req, "application/json");

httpd_resp_set_hdr(req, "Access-Control-Allow-Origin", "*");

httpd_resp_send(req, json, strlen(json));

return ESP_OK;

}

```

3.6 WORKING PRINCIPLE

The working principle of the proposed system is based on a master–slave robotic communication architecture that enables real-time control and monitoring. The system is initiated when the user sends commands through a mobile application interface. These commands are transmitted wirelessly to the master bot using a communication module such as Bluetooth or Wi-Fi.

Once the master bot receives the input signals, it processes the commands using its onboard microcontroller. The master bot acts as the central control unit and determines the appropriate actions to be performed by the slave bot. It also manages additional inputs such as push-button controls for manual operation when required.

In parallel, the camera module attached to the master bot captures real-time environmental visuals. These video signals are transmitted to the receiving unit, allowing the user to monitor the surroundings and make informed decisions during robot operation.

The processed control signals are then sent from the master bot to the slave bot through a wireless communication link. This ensures seamless and continuous data exchange between both units without physical connection.

The slave bot receives the transmitted instructions through its wireless receiver module. It interprets the commands and executes the required actions accordingly. This includes movement in different directions such as forward, backward, left, and right.

The communication between the master and slave bots is maintained through a stable wireless channel that is designed to handle continuous data exchange without interruption. This channel ensures that control signals, acknowledgments, and status updates are exchanged efficiently. Even in cases of minor interference, the system is capable of maintaining operational continuity by temporarily storing commands and executing them once the connection stabilizes.

The movement control of the slave bot is achieved through precise motor driver operations that translate digital commands into physical motion. Depending on the received instruction, the motors are activated in specific patterns to achieve forward, backward, or rotational movement.

This conversion of electrical signals into mechanical motion forms the core of the robot's working mechanism.

Overall, the system functions as an integrated unit where sensing, processing, communication, and actuation are tightly coordinated. The master bot handles intelligence and decision-making, while the slave bot performs physical actions. This division of tasks not only improves efficiency but also makes the system scalable and adaptable for more complex robotic applications in the future.

The working principle of the system is built around a tightly integrated communication and control framework that ensures seamless interaction between the user, master bot, and slave bot. The process begins when the user issues a command through the mobile application interface, which is designed to convert user inputs into structured digital signals. These signals are then transmitted through a wireless communication module such as Bluetooth or Wi-Fi to the master bot. The master bot acts as the intelligent control unit of the system, where all incoming data is received, decoded, and processed using an embedded microcontroller. During this stage, the system performs validation checks to ensure that the received instructions are accurate, complete, and free from interference. Once validated, the master bot interprets the commands and prepares them for execution by the slave bot. This layered processing approach ensures high reliability, reduces errors in transmission, and enhances overall system stability. In addition to command handling, the master bot also continuously manages communication synchronization, ensuring that the connection between devices remains active and responsive at all times without lag or data loss.

The slave bot is responsible for executing all physical actions based on the instructions received from the master bot, making it the actuation unit of the system. Once the control signals are transmitted wirelessly, the slave bot receives them through its dedicated receiver module, which decodes the signals into actionable motor instructions. These instructions are then sent to the motor driver circuit, which regulates the operation of the motors by controlling their speed, direction, and rotation patterns. This enables the robot to perform precise movements such as forward motion, backward movement, turning left or right, and stopping based on real-time commands.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 MASTER BOT

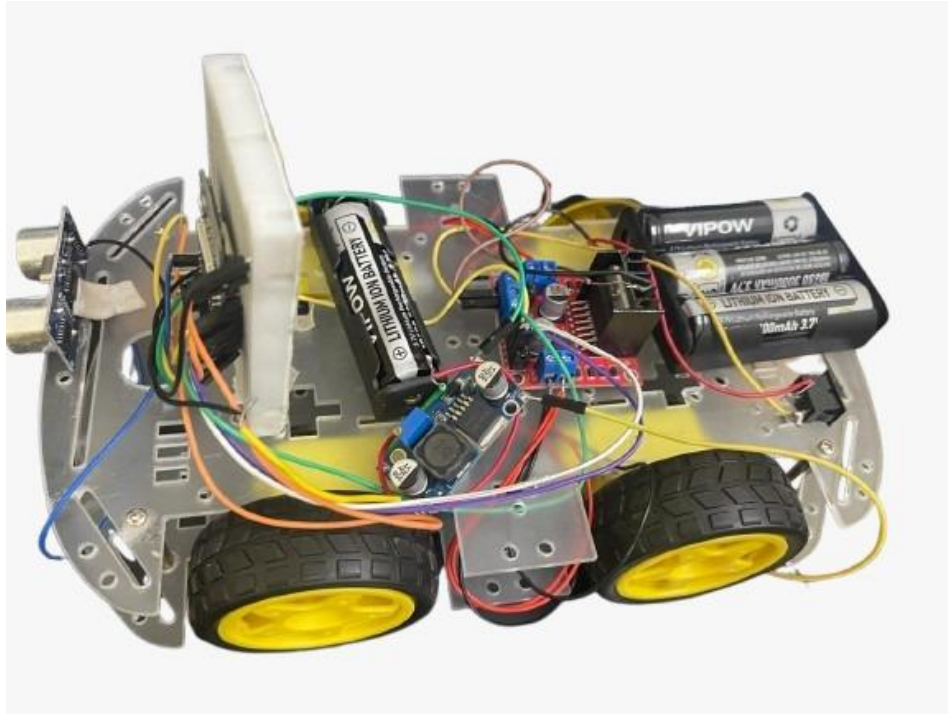


Figure 4.1 Master robot

The master bot is the central controlling unit of the swarm robotic system, responsible for processing visual data and coordinating the actions of all slave bots. It is typically equipped with an ESP32-CAM, which captures real-time images for object detection and environment analysis. Based on the processed information, the master bot sends appropriate commands to the slave bots through wireless communication, ensuring synchronized movement and efficient task execution within the system.

In addition to coordination, the master bot also plays a key role in decision-making and path planning within the system. It continuously monitors the environment and updates instructions based on dynamic conditions, such as obstacles or changes in object position. By maintaining communication with all slave bots, the master bot ensures reliability and adaptability of the swarm, making the overall system more efficient and suitable for real-time applications.

4.2 SLAVE BOT

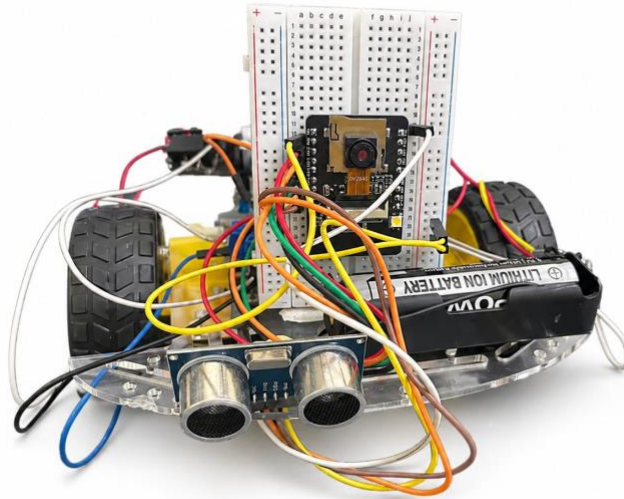


Figure 4.2: Slave robot

The slave robots are the executing units of the swarm robotic system, responsible for carrying out the instructions provided by the master bot. Each slave robot is typically built using a microcontroller such as the ESP32, along with motors and driver circuits for movement. These robots receive commands wirelessly and perform tasks such as navigation, object following, or obstacle avoidance based on the signals sent by the master bot.

Additionally, slave robots operate in a coordinated manner to achieve collective goals efficiently. They continuously communicate with the master bot and may also interact with each other to maintain proper spacing and avoid collisions. This distributed execution approach improves the scalability and flexibility of the system, allowing multiple slave robots to work together in real-time for applications like search operations, material transport, and environment monitoring.

4.3 WEB PAGE

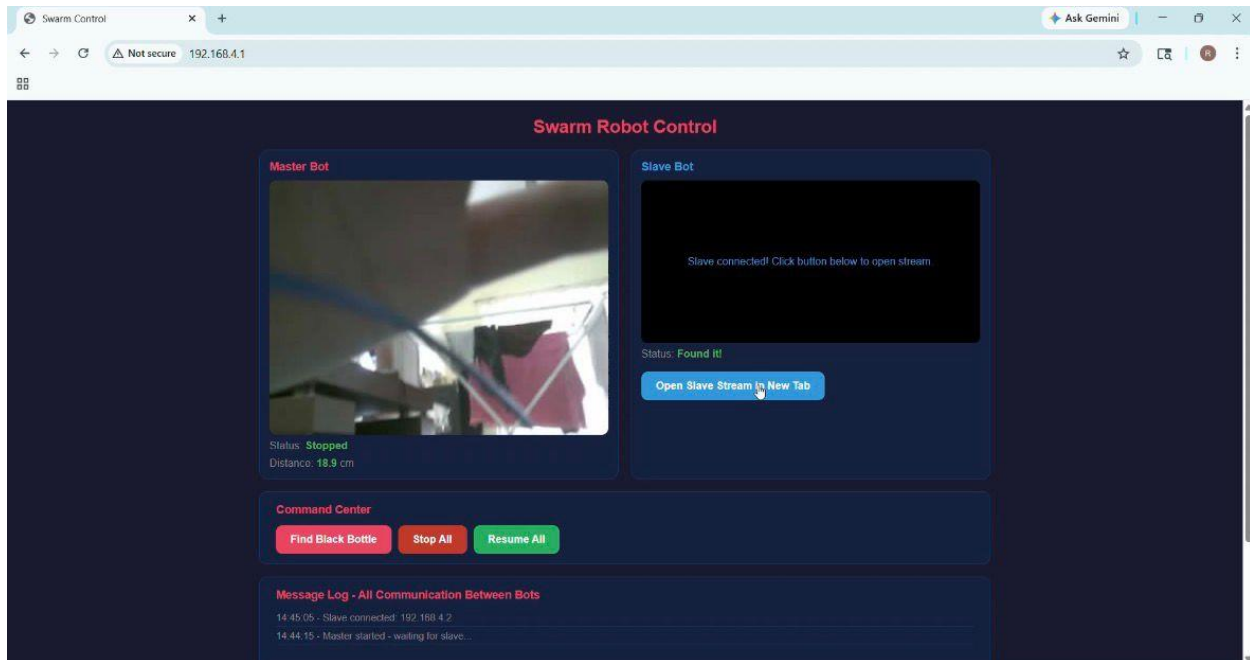


Figure 4.3: Web page of Mater Bot detection

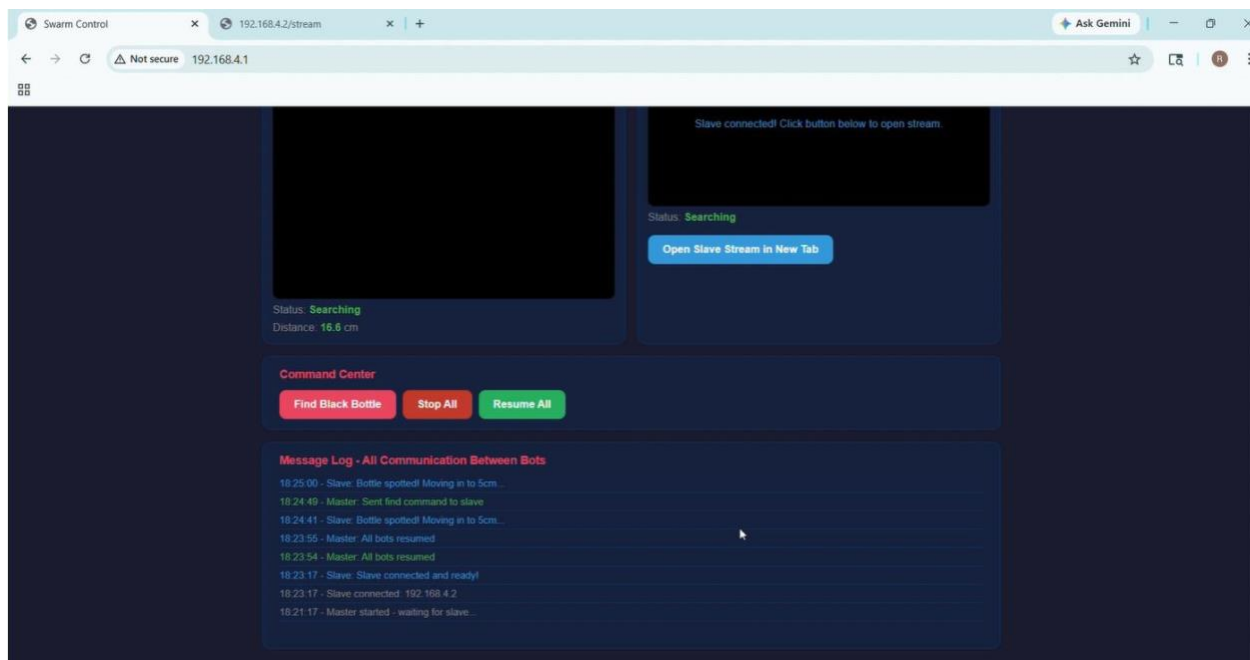


Figure 4.4: Web page - detection comments

CHAPTER 5

CONCLUSION

5.1 SUMMARY

This project successfully demonstrates the design and implementation of a cost-effective swarm robotics system using ESP32 microcontrollers integrated with computer vision-based object detection. The primary objective of enabling coordinated behavior among multiple robotic agents using a decentralized communication approach has been achieved through a master–slave architecture and Wi-Fi-based inter-robot communication.

The system effectively shows how multiple low-cost robots can work collaboratively to perform tasks such as object detection, navigation, and response execution. By integrating ESP32-CAM modules with lightweight image processing techniques, the robots are able to detect and identify objects in their environment and share relevant information with other units in the swarm. This enhances collective decision-making and allows the system to respond dynamically to environmental changes.

One of the major achievements of this project is the reduction in overall system cost while maintaining functional efficiency. Traditional swarm robotics systems often rely on expensive hardware and centralized processing units. In contrast, this implementation distributes processing among multiple ESP32 devices, reducing computational load and increasing system scalability. The use of wireless communication also eliminates the need for complex wired connections, making the system more flexible and adaptable.

The experimental results and prototype testing indicate that the swarm system is capable of maintaining coordination among multiple agents with minimal delay and acceptable accuracy in object detection and response actions. The decentralized nature of the system improves fault tolerance, meaning that even if one robot fails, the remaining units can continue functioning without complete system breakdown.

This project also highlights the practical relevance of swarm robotics in real-world applications. Potential use cases include disaster response operations where multiple robots can search and

locate survivors, environmental monitoring systems for collecting distributed data, military surveillance missions, agricultural field analysis, and industrial automation processes where coordinated multi-agent systems are required.

In addition, this work aligns with modern trends in artificial intelligence, Internet of Things (IoT), and distributed computing systems. The combination of these technologies provides a foundation for developing more intelligent, autonomous, and self-organizing robotic systems in the future.

However, the current implementation also has certain limitations. The system performance depends on network stability, processing capability of ESP32, and camera resolution constraints. Complex environments with high computational demands may require more advanced hardware or optimized algorithms to achieve better accuracy and speed.

In conclusion, this project proves that swarm robotics can be effectively implemented using low-cost embedded systems while still achieving meaningful coordination and intelligent behavior among multiple agents. The system provides a strong foundation for future enhancements such as machine learning-based decision-making, improved path planning algorithms, enhanced sensor integration, and real-time cloud connectivity for large-scale swarm coordination.

With further development, this approach has the potential to evolve into a highly efficient and scalable robotic ecosystem capable of handling complex real-world tasks in a distributed and autonomous manner.

5.2 FUTURE SCOPE

The proposed swarm robotics system using ESP32 microcontrollers and computer vision-based object detection provides a strong foundation for further research and development in the field of distributed robotics and intelligent automation. While the current implementation demonstrates basic coordination, communication, and object detection capabilities, there are several avenues through which this system can be significantly enhanced to meet real-world industrial and research-level requirements.

One of the most important directions for future improvement is the integration of advanced artificial intelligence and machine learning algorithms. By incorporating deep learning models for object recognition and decision-making, the robots can achieve higher accuracy in identifying complex objects, understanding environments, and making autonomous decisions without relying on predefined rules. Lightweight AI models optimized for edge devices can be deployed on ESP32 or upgraded microcontrollers to enable real-time intelligent processing.

The system can also be extended by integrating Internet of Things (IoT) and cloud computing platforms. By connecting swarm robots to cloud servers, large-scale data processing, remote monitoring, and centralized analytics can be achieved. This would allow real-time visualization of swarm behavior, data logging for analysis, and remote control capabilities from any location. Edge–cloud hybrid systems can further optimize performance by balancing local processing and cloud-based computation.

The slave bot is responsible for executing all physical actions based on the instructions received from the master bot, making it the actuation unit of the system. Once the control signals are transmitted wirelessly, the slave bot receives them through its dedicated receiver module, which decodes the signals into actionable motor instructions. These instructions are then sent to the motor driver circuit, which regulates the operation of the motors by controlling their speed, direction, and rotation patterns. This enables the robot to perform precise movements such as forward motion, backward movement, turning left or right, and stopping based on real-time commands.

Another promising direction is the development of fully autonomous navigation and path planning systems. By implementing algorithms such as A*, Dijkstra's algorithm, or reinforcement learning-based navigation, robots can independently explore unknown environments, avoid obstacles more efficiently, and optimize their movement paths for better performance in dynamic conditions.

The system also has potential applications in real-world industrial and societal domains, which can be explored in future work. These include disaster management operations where swarm robots can assist in search and rescue missions, agricultural automation for crop monitoring and spraying, military surveillance systems for area monitoring, warehouse automation for logistics

management, and environmental monitoring for pollution tracking and data collection across large geographical areas.

Furthermore, future enhancements can focus on improving energy efficiency and battery management systems. Since swarm robots often operate in remote or large-scale environments, optimizing power consumption through intelligent sleep modes, energy harvesting techniques, and efficient task scheduling will be crucial for long-duration missions.

Finally, the system can evolve into a fully adaptive and self-learning robotic ecosystem capable of continuous improvement over time. By combining AI, IoT, distributed computing, and advanced robotics principles, future swarm systems can become highly intelligent, scalable, and capable of handling complex tasks with minimal human intervention.

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